

Assignment 4

Artificial Intelligence

Q1. State Space Representation (Case Study)

A logistics company is trying to optimize its delivery system in a smart city. The city is represented as a graph where nodes represent locations and edges represent routes with associated costs (distance, traffic, fuel consumption). The system must determine the most efficient route for delivery vehicles while considering constraints like traffic congestion and delivery deadlines.

Tasks:

- Model the problem using **state space representation** (define state, operators, initial state, goal state).
- Explain how **search algorithms** like Breadth-First Search, Depth-First Search, and A* can be applied.
- Compare uninformed vs informed search strategies in this context.
- Suggest the most suitable approach and justify your answer.

Q2. Reinforcement Learning (Case Study)

A self-driving car company is developing an AI agent to learn how to drive safely in urban environments. The car receives rewards for safe driving and penalties for collisions, traffic violations, or inefficient routes.

Tasks:

- Formulate the problem as a **Reinforcement Learning (RL)** model (states, actions, rewards, environment).
- Explain how **Q-learning** or **Deep Reinforcement Learning** can be used.
- Discuss the exploration vs exploitation trade-off in this scenario.
- Identify challenges such as sparse rewards, safety concerns, and real-world deployment.

Q3. Neural Network Learning & Genetic Algorithms (Case Study)

A healthcare organization wants to develop a predictive system for early detection of heart disease. They are considering two approaches: Neural Networks and Genetic Algorithms.

Tasks:

- Explain how a **Neural Network model** can be designed for prediction (architecture, training, backpropagation).
- Describe how **Genetic Algorithms (GA)** can optimize feature selection or model parameters.
- Compare both approaches in terms of accuracy, interpretability, and computational cost.
- Suggest a hybrid approach combining Neural Networks and Genetic Algorithms.

Q4. Natural Language Processing (NLP) (Case Study)

A multinational company wants to build a chatbot that can handle customer queries in multiple languages and understand user sentiment.

Tasks:

- Explain how NLP techniques like **tokenization, stemming, and parsing** can be used.
- Describe the role of **language models and transformers** in chatbot development.

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- Discuss challenges like ambiguity, sarcasm, and multilingual processing.
- Propose a system architecture for the chatbot.

Q5. Machine Learning (ML) (Case Study)

An e-commerce company wants to improve its recommendation system to increase sales and customer satisfaction.

Tasks:

- Identify the type of ML approach (supervised, unsupervised, reinforcement learning).
- Explain how **collaborative filtering and content-based filtering** can be applied.
- Discuss data preprocessing, feature engineering, and evaluation metrics.